1 Closure Properties

Closure Properties

- Recall that we can carry out operations on one or more languages to obtain a new language
- Very useful in studying the properties of one language by relating it to other (better understood) languages
- Most useful when the operations are sophisticated, yet are guaranteed to preserve interesting properties of the language.
- Today: A variety of operations which preserve regularity
  - i.e., the universe of regular languages is *closed* under these operations

Definition 1. Regular Languages are closed under an operation op on languages if

$$L_1, L_2, \ldots L_n \text{ regular } \implies L = \text{op}(L_1, L_2, \ldots L_n) \text{ is regular}$$

1.1 Boolean Operators

Operations from Regular Expressions

Proposition 2. *Regular Languages are closed under* $\cup$*,* $\circ$ *and* **.*

Proof. (Summarizing previous arguments.)

- L_1, L_2 regular $\implies \exists$ regexes R_1, R_2 s.t. $L_1 = \mathbf{L}(R_1)$ and $L_2 = \mathbf{L}(R_2)$.
  - $\implies L_1 \cup L_2 = \mathbf{L}(R_1 \cup R_2) \implies L_1 \cup L_2$ regular.
  - $\implies L_1 \circ L_2 = \mathbf{L}(R_1 \circ R_2) \implies L_1 \circ L_2$ regular.
  - $\implies L_1^* = \mathbf{L}(R_1^*) \implies L_1^*$ regular. □

Closure Under Complementation

Proposition 3. *Regular Languages are closed under complementation, i.e., if* L *is regular then* $\overline{L} = \Sigma^* \setminus L$ *is also regular.*

Proof.
- If L is regular, then there is a DFA $M = (Q, \Sigma, \delta, q_0, F)$ such that $L = L(M)$.
- Then, $\overline{M} = (Q, \Sigma, \delta, q_0, Q \setminus F)$ (i.e., switch accept and non-accept states) accepts $\overline{L}$. □

What happens if M (above) was an *NFA*?

Closure under $\cap$

Proposition 4. *Regular Languages are closed under intersection, i.e., if L_1 and L_2 are regular then $L_1 \cap L_2$ is also regular.*

Proof. Observe that $L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}}$. Since regular languages are closed under union and complementation, we have

- $\overline{L_1}$ and $\overline{L_2}$ are regular
- $\overline{L_1} \cup \overline{L_2}$ is regular
- Hence, $L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}}$ is regular. □

Is there a direct proof for intersection (yielding a smaller DFA)? ______

Cross-Product Construction

Let $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ and $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ be DFAs recognizing L_1 and L_2, respectively.

Idea: Run M_1 and M_2 in parallel on the same input and accept if both M_1 and M_2 accept.

Consider $M = (Q, \Sigma, \delta, q_0, F)$ defined as follows

- $Q = Q_1 \times Q_2$
- $q_0 = \langle q_1, q_2 \rangle$
- $\delta(\langle p_1, p_2 \rangle, a) = \langle \delta_1(p_1, a), \delta_2(p_2, a) \rangle$
- $F = F_1 \times F_2$

M accepts $L_1 \cap L_2$ (exercise)

What happens if M_1 and M_2 where NFAs? Still works! Set $\delta(\langle p_1, p_2 \rangle, a) = \delta_1(p_1, a) \times \delta_2(p_2, a)$.

An Example